Natural Language Understanding part LAB 5

Christian Dal Farra (mat. 258243)

University of Trento

christian.dalfarra@studenti.unitn.it

1. Introduction

In this work, I focused on the joint problem of intent classification and slot filling for language understanding using the ATIS dataset. I began with a baseline LSTM-based model and gradually improved it by adding bidirectional layers and dropout, carefully tuning the hyperparameters to achieve better performance. I then moved to a transformer-based approach by fine-tuning a pre-trained BERT model in a multi-task setting. Particular attention was paid to managing subword tokenization when predicting slot labels.

2. Implementation details

Starting from the code given in the Lab 5, i set up experiments with 5 runs on 200 epochs for each with Adam optimizer and learning rate of 0.0001, 200 hidden size and 300 of embedding size, the loss function was cross entropy, evaluate on Slots F1 and intent accuracy, first baseline LSTM performed good with this configuration, then i added the bidirectionality on the LSTM module, changing the parameter `bidirectional` to True in the LSTM module, and get better performances, so I added the last requirement for the part A that is dropout, one dropout after the embeddings and one before the last layer in both the probability was 0.5, also with this modification i got the best performances for the part A.

In Part B of this project, I replaced the LSTM encoder with a pre-trained BERT model, following the joint intent classification and slot filling approach proposed by Chen et al [1]. The main motivation was to take advantage of BERT's contextualized representations, which are able to model semantic information more effectively than recurrent architectures. In practice, I fine-tuned the "bert-base-uncased" model in a multi-task setting, using the representation of the special `[CLS]` token for intent classification and the token-level hidden states for slot filling.

Introducing BERT poses some challenges, mainly related to tokenization, unlike standard word-level tokenizers, BERT relies on WordPiece tokenization, which can split a single word into multiple sub-tokens. Since slot labels in the ATIS dataset are defined at the word level, this creates alignment issues between the tokenized input and the corresponding slot annotations. For instance, a the word "tokenization" is broken into multiple sub-tokens ['token', '##ization'], resulting in a mismatch between the number of tokens and the number of slot labels.

To handle this problem, custom data collation and loss-masking strategies. Each utterance was tokenized using `AutoTokenizer`, which provides a mapping between sub-tokens and their original words. This made it possible to assign the correct slot label only to the first sub-token of each word, while all remaining sub-tokens were given a `PAD_TOKEN` that is ignored during training. Additional padding was added to account for BERT's special tokens, `[CLS]` and `[SEP]`, as well as to ensure uniform sequence lengths using `[PAD]` tokens. Attention masks were used to prevent padded tokens from influencing the model's representations.

During the evaluation, both predicted and ground truth slot labels were post-processed to remove padding and special tokens, ensuring proper alignment before computing the F1 score. Overall, this strategy allowed for a stable and effective fine-tuning of BERT for the joint intent classification and slot filling task on the ATIS dataset.

In this case the training were done using single run, over 50 epoch as referenced in the paper [1], I try different learning rates and 0.00005 gives the best performances, I have also tried to add a dropout layer with probability 0.2 but did not achieve better results, so I did not include it in the table.

3. Results

In Part A and B the model performance is evaluated using intent classification accuracy and the CoNLL F1 score, using the script given in lab, on the ATIS dataset. In the Table 1 are shown the results for the Part A, each incremental change achieves better results compared to the previous ones, in Figure 1 there is the loss plot for the best experiment of part A.

Model	Slot F1	Intent acc
LSTM	0.923 ± 0.002	0.935 ± 0.006
LSTM Bidir	0.936 ± 0.004	0.94 ± 0.005
LSTM BiDir Dropout	0.942 ± 0.003	0.95 ± 0.004

Table 1: *Part A results*

In the Table 2 are shown the result for the Part B, related to BERT, that achieve the best results for this task, in Figure 2 there is the loss plot for BERT fine-tuning.

Model	Slot F1	Intent acc
BERT	0.953	0.978

Table 2: *Part B results*

4. References

[1] Q. Chen, Z. Zhuo, and W. Wang, "Bert for joint intent classification and slot filling," 2019. [Online]. Available: https://arxiv.org/abs/1902.10909

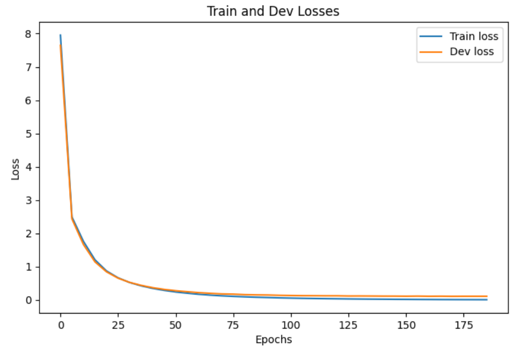

Figure 1: *Loss plot for the last experiment of the first part*

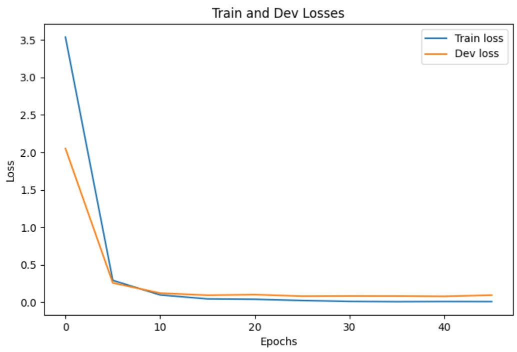

Figure 2: *Loss plot for BERT*